

Hauke Sandhaus

HCI RESEARCHER
& UX TECHNOLOGIST

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Designing for user and worker autonomy, imagining AI systems that not just respect but support human autonomy. Dissertation work develops theory that distinguishes legitimate machine influence from manipulation.

CURRENTLY

Cornell Tech / Ph.D. Candidate in Information Science

STARTED AUGUST 2021 · EXPECTED GRADUATION JULY 2027 · NYC, NY, USA

Advised by Helen Nissenbaum (Chair), Wendy Ju (Co-Chair), and Qian Yang (Committee Member).

PROFESSIONAL EXPERIENCE

Cornell Tech / Graduate Research Assistant — Manipulative AI

SEPTEMBER 2026 – PRESENT

With Helen Nissenbaum. Empirical studies on the minimum signals AI systems can send to influence human behavior. Draws inspiration from manipulative-signaling studies in animal behavior research.

Center for Responsible AI at NYU / Research Assistant

MARCH 2023 – DECEMBER 2023

Worked on the 'University of Notre Dame - IBM' Technology Ethics Lab Grant Auditing AI Initiative on the project 'Expanding AI Audits to Include Instruments: Accountability, Measurements, and Data in Motion Capture Technology.'

- Collaborated with researchers from University of Michigan, Intel Labs, and New York University on an award-winning CHI publication
- Investigated bias in measurement standards for AI hardware
- Contributed to frameworks for auditing AI technologies

Cornell Tech / Graduate Research Assistant — Designer Decision-Making

JANUARY 2023 – MAY 2023

With Qian Yang. One-semester grant on external pressures shaping designer decision-making, with a mental-health-technology focus.

Cornell Tech / Graduate Research Assistant — Human-AV Interaction

JANUARY 2022 – AUGUST 2023

With Qian Yang. On-road studies of human-autonomous vehicle interaction and analysis of safety-critical data-sharing practices among AV companies. Fed into CSCW '25 and AutoUI/IEEE ITSC '24 publications.

Volkswagen Group Future Center / UX Technologist

OCTOBER 2018 – AUGUST 2021

Led the design of inclusive user experiences for fully autonomous vehicles and developed next-generation driver interfaces. Involved in the creation and transformation of CARIAD.

- Improved User Experience Questionnaire (UEQ) benchmarks by 30% over state-of-the-art through inclusive design approaches
- Developed and implemented a next-generation digital cockpit that improved accessibility of steering wheel functions without affecting driving performance
- Created high-fidelity AI prototypes using cutting-edge research papers and custom computer vision pipelines that were adopted for production development
- Conducted on-road studies for ambient in-cabin lighting systems that identified critical safety, comfort, and motion sickness factors for AV passengers
- Designed and prototyped LLM-based digital assistant interfaces for in-vehicle use, now in final stages before market release
- Established GDPR-compliant research practices that were adopted across the VW Group
- Successfully transferred multiple vision projects to strategy teams, with implementations adopted by Porsche, Skoda, and Audi

Robert Bosch / Intern & External Master Theses

AUGUST 2017 – JANUARY 2018

Designed V2X interfaces and researched interfaces for autonomous vehicles tailored for older adults and disabled users.

- Conducted research on autonomous vehicle interfaces for older adults and people with disabilities
- Designed Vehicle-to-Everything (V2X) interfaces for future mobility solutions

ELIXIER / Student Researcher

AUGUST 2016 – APRIL 2017

Conducted usability testing of multimodal and augmented workbenches for educational purposes and developed an AR manual for chemistry experiments.

100%FAT / External Bachelor Theses

JANUARY 2015 – MAY 2015

Designed, developed, and evaluated a prototype for a touchless interactive showcase in the DesignLab Twente. The prototype was permanently installed in the Design Hall.

Twente Academy / Student Teacher

MARCH 2014 – SEPTEMBER 2014

Taught upper secondary school pupils programming using the Arduino microprocessor.

EDUCATION

Cornell University, Ann S. Bowers College of Computing and Information Science / M.Sc.

2021 – 2024

Information Science

Passed Qualification Exam on October 2nd, 2024

- A Exam Title: 'Navigating Ethical Design Under External Pressures'
- Dissertation Topic: 'Empowering technology designers to create ethical technologies and resist external pressures.'

Bauhaus Universität Weimar / M.Sc.

2015 – 2018

Human-Computer Interaction

Graduated with honors (GPA 3.9)

- Master Thesis: 'An Ambient Display for Fully Autonomous Vehicles'
- Best Thesis of the Year
- Publication: 'A WOZ Study of Feedforward Information on an Ambient Display in Autonomous Cars' in The 31st Annual ACM Symposium on User Interface Software and Technology

Universiteit Twente / B.Sc.

2011 – 2015

Creative Technology

Cum Laude (Thesis GPA 4.0)

- Bachelor Thesis: 'Development of a 3D Navigable Interface for a Touchless Showcase'
- Represented graduation year and gave graduation speech
- Prototype permanently installed in the Design Hall

Czech Technical University / Semester Abroad

2013 – 2014

Electrical Engineering and Telecommunications

Gymnasium im Schloss / Abitur

2003 – 2011

Science Branch (STEM)

- Graduated with majors in Art & Religion
- Participated in and won science competitions

SERVICE & TEACHING

Hybrid Chair — CHIWORK 2026

FALL 2025 – SUMMER 2026 · ZURICH, SWITZERLAND

Chairing hybrid participation and infrastructure for the 5th Annual Symposium on Human-Computer Interaction for Work.

Instructor, Ethical Vibe Coding — Cornell Tech

SUMMER 2026 · NEW YORK CITY

Three-week course on ethical AI-assisted coding for high school students at Cornell Tech's Summer Innovation Intensives. With Jonathan Segal.

Co-host NYC Privacy Day — Cornell Tech

FALL 2025 · NEW YORK CITY

Managed the website, sign-ups, and the poster and demo track for NYC Privacy Day, an initiative between privacy research institutes in the NYC area.

Instructor Ethics for Public Interest Technology — Cornell Tech

SUMMER 2025 · NEW YORK CITY

Co-instructed ethics workshop on conscientious technology design, developing materials, and leading activities on value-driven design thinking (pitechethics.github.io). With Helen Nissenbaum.

Teaching Assistant in Privacy in the Digital Age — Cornell Tech

FALL 2024 · NEW YORK CITY

Head TA in master level privacy class. Led assignments, in-class privacy policy exercise and a lecture on usable privacy. With Helen Nissenbaum and Vitaly Shmatikov.

Teaching Assistant in UX & User Research — Cornell Tech

SPRING 2026 · NEW YORK CITY

Assisted in teaching a master's level course focused on the UX lifecycle and user research methods. With Shiri Azenkot.

Teaching Assistant in Interactive Device Design — Cornell Tech

FALL 2023 – FALL 2025 · NEW YORK CITY

Head TA in master level prototyping class. Maintained project code repository, labs, and oversaw grading. Updated the class for AI-codesign of technology. With Wendy Ju.

ISGSA Liaison — Cornell University

FALL 2022 – FALL 2023 · NEW YORK CITY

Information Science Graduate Student Association Liaison for NYC to Ithaca campus collaboration.

Teaching Assistant in HCI Design — Cornell University

FALL 2021 · ITHACA

Assisted in teaching undergraduate and master level course in human computer interaction. Oversaw grading and individual sections. With Gilly Leshed.

Member — We Drive Proud

2019 – 2021 · POTSDAM, GERMANY

Member of the VW network to advocate being out at work.

Reviewer — Peer Review

2018 – ONGOING

Peer reviewer at FAccT, CHI, DIS, AutoUI, CSCW, MuC, and CHIwork.

FELLOWSHIPS & AWARDS

Public Interest Responsible Technology Fund

JUNE 2026 · CORNELL TECH

Teaching grant to instruct a PhD-fellow cohort on value-centered design methods.

DLI Doctoral Fellow

2023-2024 · CORNELL TECH

The Digital Life Initiative supports examining the complexities and tensions present in digital technologies involving ethics, policy, politics, and quality of life.

Siegel Public Interest Technology Impact Fellowship

MARCH 2021 · CORNELL TECH

Funded to work for not-for-profit organization YAI. YAI is one of the largest organizations in the U.S. providing children and adults with intellectual and developmental disabilities (I/DD) a comprehensive range of services.

Thesis of the Year

JULY 2015 · UNIVERSITEIT TWENTE

Selected by jury to speak at graduation ceremony. Bachelor thesis presented on behalf of the creative technology program. In 2026, he won the prestigious \$5,000 NYC Privacy Day Award for his groundbreaking contributions to resistance techniques through LLM obfuscation.

PUBLICATIONS

- Kashif Imteyaz, Mohammad Rashidujjaman Rifat, Divya Ramesh, Steven R Rick, Simo Hosio, **Hauke Sandhaus**, Advait Sarkar, Christoph Riedl, Saiph Savage (November 2026). "Nobody Did This": Contribution, Originality, and Accountability in Agent-Mediated Collaboration. *Companion Publication of the 2026 ACM Conference on Computer-Supported Cooperative Work and Social Computing*. · Workshop co-organizer
- **Hauke Sandhaus**, Doris Maria Rhomberg, Helen Nissenbaum (June 2026). Making Indecent Persuasion Visible: How Evaluation Metrics Shape UX Designers' Ethical Reasoning. *CHIWORK '26: Proceedings of the 5th Annual Symposium on Human-Computer Interaction for Work (Late-Breaking Work)*.
- **Hauke Sandhaus**, Kashif Imteyaz, Mohammed Almutairi, Pooja Prajod, Divya Ramesh, Saiph Savage, Qian Yang, Michael Muller (June 2026). Interrogating GenAI Augmentation for CHIworkers: Strategies for Professional Autonomy and Accountability. *CHIWORK '26: Proceedings of the 5th Annual Symposium on Human-Computer Interaction for Work (Workshop)*. · Workshop co-organizer
- Maria Teresa Parreira, Pranav Prabhat Sinha, **Hauke Sandhaus**, Wendy Ju (June 2026). Visible Adoption, Untracked Contribution: GitHub Evidence of the Accountability Gap Across Three Cohorts of an HCI Prototyping Course. *CHIWORK '26: Adjunct Proceedings of the 5th Annual Symposium on Human-Computer Interaction for Work*. · Workshop paper at CHIWORK '26 (workshop I co-organized)
- Thomas Ebmeyer, Johanna Gunawan, Colin M Gray, Lorena Sánchez Chamorro, Alberto Monge Roffarello, Katie Seaborn, **Hauke Sandhaus**, Kai Lukoff, Gian-Luca Savino, Shruthi Sai Chivukula, Satoshi Nakamura (April 2026). Bridge Over Troubled Water: Aligning Commercial Incentives With Ethical Design Practice To Combat Deceptive Patterns. *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*. · Workshop co-organizer
- **Hauke Sandhaus**, Doris Maria Rhomberg (2026). Deception Is Ugly: Linking Aesthetic Judgment to Perceived Manipulation in Dark Patterns. *Zeitschrift für Psychologie*. · Accepted, forthcoming
- **Hauke Sandhaus**, Angel Hsing-Chi Hwang, Wendy Ju, Qian Yang (October 2025). My Precious Crash Data: Barriers and Opportunities in Encouraging Autonomous Driving Companies to Share Safety-Critical Data. *Proceedings of the ACM on Human-Computer Interaction*.
- Matt Franchi*, **Hauke Sandhaus***, Madiha Zahrah Choksi, Severin Engelmann, Wendy Ju, Helen Nissenbaum (June 2025). Privacy of Groups in Dense Street Imagery. *Proceedings of*

the 2025 ACM Conference on Fairness, Accountability, and Transparency. · * = equal contribution

- **Hauke Sandhaus**, Qiuquan Gu, Maria Teresa Parreira, Wendy Ju (July 2025). Co-Designing with Transformers: Unpacking the Complex Role of GenAI in Interactive System Design Education. *Proceedings of the 2025 ACM Designing Interactive Systems Conference*.
- Rachel Di Pirro, **Hauke Sandhaus**, David Goedicke, Dan Calderone, Meeko Oishi, Wendy Ju (September 2024). Characterizing Cultural Differences in Naturalistic Driving Interactions. *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*.
- **Hauke Sandhaus**, Maria Teresa Parreira, Wendy Ju (May 2024). Student Reflections on Self-Initiated GenAI Use in HCI Education. *LLMs as Research Tools Workshop at the CHI Conference on Human Factors in Computing Systems*.
- Doris Maria Rhomberg*, **Hauke Sandhaus*** (May 2024). Towards Quantifying Ethical User Experience: Evaluating User Perceptions of Dark Patterns in Social Media. *Mobilizing Research and Regulatory Action on Dark Patterns and Deceptive Design Practices Workshop at the CHI Conference on Human Factors in Computing Systems*. · * = equal contribution
- Emma Harvey, **Hauke Sandhaus**, Abigail Z Jacobs, Emanuel Moss, Mona Sloane (January 2024). The Cadaver in the Machine: The Social Practices of Measurement and Validation in Motion Capture Technology. *Proceedings of the CHI Conference on Human Factors in Computing Systems*.
- Navit Klein, **Hauke Sandhaus**, David Goedicke, Wendy Ju, Avi Parush (2024). Modeling Social Situation Awareness in Driving Interactions. *Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*.
- Patrick Ebel, Pavlo Bazilinskyy, Mark Colley, Courtney Michael Goodridge, Philipp Hock, Christian P. Janssen, **Hauke Sandhaus**, Aravinda Ramakrishnan Srinivasan, Philipp Wintersberger (2024). Changing Lanes Toward Open Science: Openness and Transparency in Automotive User Research. *Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*.
- **Hauke Sandhaus**, Madiha Zahrah Choksi, Wendy Ju (2024). Regaining Trust: Impact of Transparent User Interface Design on Acceptance of Camera-Based In-Car Health Monitoring Systems. *Adjunct Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*.
- Jieun Kim*, **Hauke Sandhaus***, Susan R Fussell (October 2023). VR Job Interview Using a Gender-Swapped Avatar. *Computer Supported Cooperative Work and Social Computing*. · * = equal contribution
- **Hauke Sandhaus** (April 2023). Promoting Bright Patterns. *CHI '23 Workshop: Designing Technology and Policy Simultaneously*. · This paper started the brightpatterns.org website
- **Hauke Sandhaus**, Wendy Ju, Qian Yang (April 2023). Towards Prototyping Driverless Vehicle Behaviors, City Design, and Policies Simultaneously. *CHI '23 Workshop: Designing Technology and Policy Simultaneously*.
- **Hauke Sandhaus**, Eva Hornecker (October 2018). A WOZ Study of Feedforward Information on an Ambient Display in Autonomous Cars. *The 31st Annual ACM Symposium on User Interface Software and Technology Adjunct Proceedings*.
- Aline Martinez, Michaela Honauer, **Hauke Sandhaus**, Eva Hornecker (2018). Smart Textiles in the Performing Arts. *Textiles, Identity and Innovation: Design the Future*.
- **Hauke Sandhaus** (2015). Development of a 3D Navigable Interface for a Touchless Showcase. · Bachelor's thesis, University of Twente

SKILLS

Coding: Python, Java, JavaScript, React.js, Node.js, Android, Unity3D, C#, Arduino, Open Frameworks

AI: fastai, Pinecone, Teachable Machines, TensorFlow, ml5.js, OpenCV, Scikit-learn, DepthAI, LLM integration, Custom computer vision pipelines

Data Analytics: Python data visualization, Tableau, R, Matlab, d3.js

Design: Interface design, Lottie, AE, 3D modeling, PSD, Framer

User Research: Contextual & expert interviews, Ethnography, Surveys, Usability testing, On-road studies

UX: Personas, Wireframes, Focus groups, Human factors, Inclusive design, Accessibility

Hardware: CAD, Rapid Prototyping, 3D manufacturing

Management: Team leadership, Conflict resolution, Strategy, Scaled agile methods (certified), UX strategy